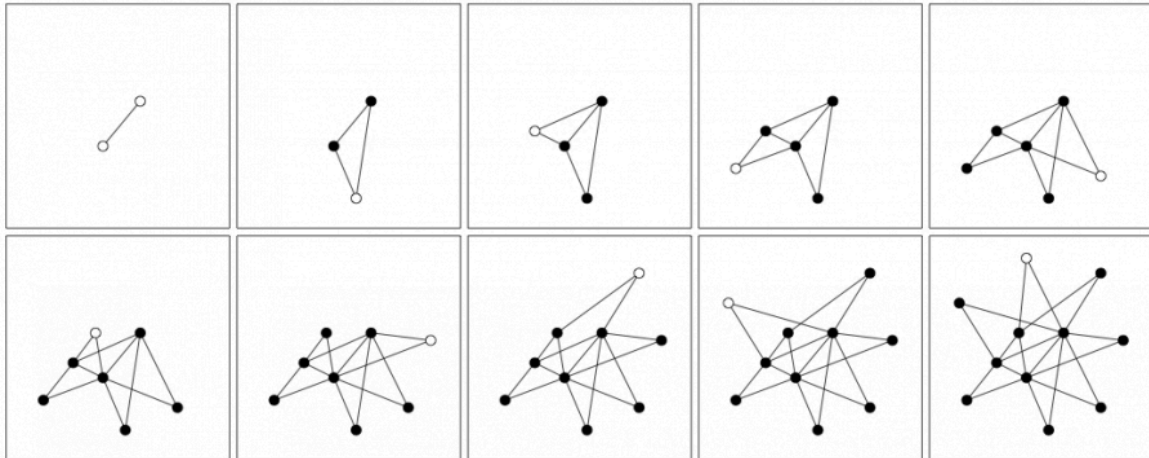


L4: Preferential Attachment Model

So long as the sum of the degrees is even, the configuration model can generate networks with any degree distribution – including power-laws with any exponent. However, the configuration model does not suggest any **“generating mechanism”** that explains how a network can gradually acquire a power-law degree distribution.



Evolution of the Barabási-Albert Model, Figure 5.3 from networksciencebook.com by Albert-László Barabási

One such generating model is known as **“preferential attachment”** (or *PA model* or *“Barabási-Albert” model*). In this model, the network grows by one node at each time step – so the network has t nodes after t time steps. Every time a new node is added to the network, it connects to m existing nodes (m is the same for all new nodes – suppose that we allow self-loops and multi-edges for now). The neighbors of the new node are chosen randomly but with a non-uniform probability, as follows.

Suppose that the new node arrives at a point in time t , and let $k_i(t)$ be the degree of node i at that time. The probability that the new node will connect to node i is:

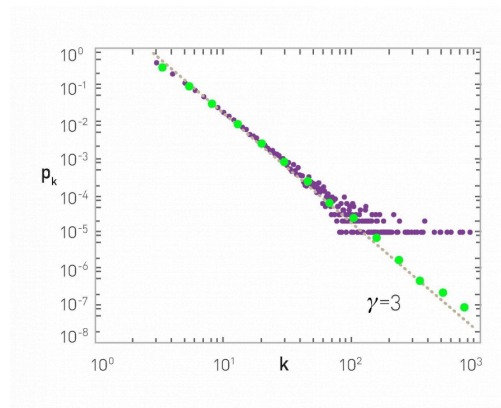
$$\Pi_i(t) = \frac{k_i(t)}{\sum_{j=1}^t k_j(t)} = \frac{k_i(t)}{2mt}$$

In other words, in the preferential attachment model, the network grows over time and new nodes are more likely to connect to nodes with higher degrees (see above plot). This is a **“rich get richer”** effect because nodes with higher degree attract more connections from new nodes, making their degree even higher relative to other nodes.

Later in the course, we will return to this model and study its behavior mathematically.

For now, we only mention without proof that this model produces power-law networks with an exponent $\alpha=3$. The value of the parameter m (number of edges of new node) does not affect the exponent of the distribution.

The degree distribution at the plot below refers to a network that was created with the preferential attachment model, after generating $n=100,000$ nodes and with $m=3$ (the green dots show a log-binned estimate of the distribution while the purple dots show a linearly-binned histogram).



The Degree Distribution, Figure 5.4 from networksciencebook.com by Albert-László Barabási

The main value of the preferential attachment model is that it suggests that power-law networks can be generated through the combined effect of two mechanisms: growth and preferential connecting to nodes with a higher degree. Either of these two mechanisms on its own would not be sufficient to produce power-law networks.

Food for Thought

How would you modify the preferential attachment model so that you get an exponent α between 2 and 3?